

Extrema and the Mean Value Theorem



Critical points and extrema

- 1 Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.
 - 2 Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.
-

The Extreme Value Theorem

- 3 State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.
-

Rolle's Theorem

- 4 Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.
 - 5 Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.
-

The Mean Value Theorem

- 6 For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.
 - 7 A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.
-

Using the MVT to bound function values

- 8 Given $f(2) = 5$ and $-3 \leq f'(x) \leq 4$ for all x , use the MVT to find the range of possible values for $f(6)$.
-

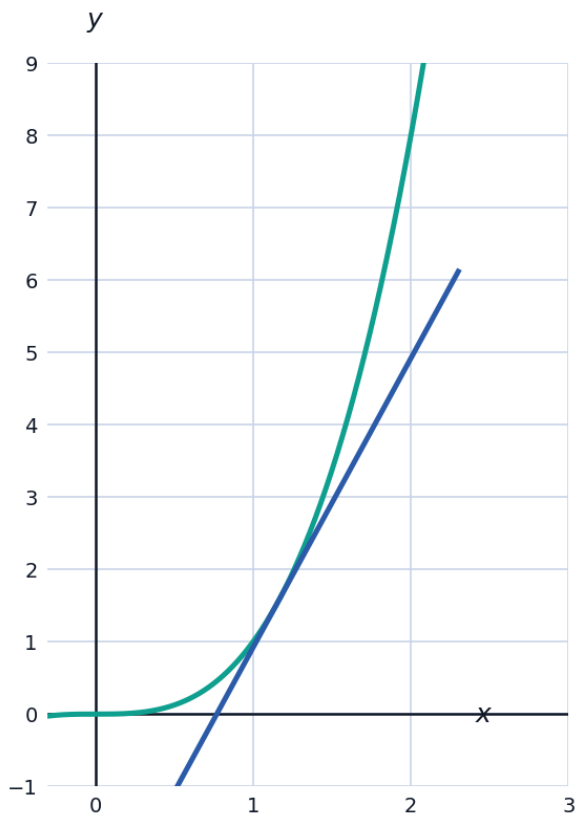
Consequences of the MVT: monotonicity

- 9 Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.

- 10 Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.
-

Visualizing the Mean Value Theorem

- 11 The graph shows $f(x) = x^3$ on $[0, 2]$ together with the secant line through the endpoints and the tangent line at $x = \frac{2}{\sqrt{3}}$ (found earlier).



Absolute extrema on a closed interval — a capstone problem

- 12 Find the absolute maximum and minimum of $f(x) = x^4 - 4x^2 + 3$ on $[-2, 2]$, checking all critical points and endpoints.
-

Rolle's Theorem with trigonometric functions

- 13 Verify that Rolle's Theorem applies to $f(x) = \sin x$ on $[0, \pi]$, and find the value of c guaranteed by the theorem.
-

An applied Mean Value Theorem problem

- 14 A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.